

2.a

$$J(\theta) = -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} \log(h_{\theta}(x^{(i)})) + (1-y^{(i)}) \log(1-h_{\theta}(x^{(i)})) \right]$$

where $y^{(i)} \in \{0, 1\}$, $h_{\theta}(x) = g(\theta^T x)$, $g(z) = \frac{1}{1+e^{-z}}$

To find Hessian Matrix of $J(\theta)$, ie $\nabla^2 J(\theta)$, Let's start with finding the first order partial derivative first.

$$J(\theta) = -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} \log(h_{\theta}(x^{(i)})) + (1-y^{(i)}) \log(1-h_{\theta}(x^{(i)})) \right]$$

Let $T_1(\theta) = \log(h_{\theta}(x^{(i)}))$

$$\begin{aligned} \frac{\partial T_1(\theta)}{\partial \theta_j} &= \frac{\partial}{\partial \theta_j} (\log(h_{\theta}(x^{(i)}))) = \frac{\partial}{\partial \theta_j} (\log(g(\theta^T x))) \\ &= \frac{1}{g(\theta^T x)} \cdot g'(\theta^T x) \cdot x_j \end{aligned}$$

We know $g'(z) = g(z) \cdot (1-g(z))$, using this

$$\frac{\partial T_1(\theta)}{\partial \theta_j} = \frac{1}{g(\theta^T x)} \cdot g(\theta^T x) (1-g(\theta^T x)) \cdot x_j \quad \text{--- (1)}$$

$$\boxed{\frac{\partial T_1(\theta)}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} (\log(h_{\theta}(x^{(i)}))) = 1 - g(\theta^T x) \cdot x_j}$$

Let $T_2(\theta) = \log(1-h_{\theta}(x^{(i)}))$

$$\begin{aligned} \frac{\partial T_2(\theta)}{\partial \theta_j} &= \frac{\partial}{\partial \theta_j} (\log(1-h_{\theta}(x^{(i)}))) = \frac{\partial}{\partial \theta_j} (\log(1-g(\theta^T x))) \\ &= \frac{\partial}{\partial \theta_j} (\log(1-g(\theta^T x))) = \frac{1}{1-g(\theta^T x)} \cdot -g'(\theta^T x) \cdot x_j \end{aligned}$$

$$\frac{\partial T_2(\theta)}{\partial \theta_j} = \frac{-g'(\theta^T x) x_j}{1 - g(\theta^T x)} = \frac{-g(\theta^T x) \cdot (1 - g(\theta^T x))}{(1 - g(\theta^T x))} \cdot x_j$$

$$\therefore \frac{\partial T_2(\theta)}{\partial \theta_j} = \frac{\partial \log(1 - h_\theta(x^{(i)}))}{\partial \theta_j} = -g(\theta^T x) \cdot x_j \quad \text{--- (2)}$$

plugging (1) and (2) in $\frac{\partial J}{\partial \theta_j}$

$$\frac{\partial J}{\partial \theta_j} = -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} [1 - g(\theta^T x^{(i)})] x_j^{(i)} - (1 - y^{(i)}) g(\theta^T x^{(i)}) x_j^{(i)} \right]$$

$$= -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} [1 - g(\theta^T x^{(i)})] \cdot x_j^{(i)} - (1 - y^{(i)}) g(\theta^T x^{(i)}) x_j^{(i)} \right]$$

$$= -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} - y^{(i)} g(\theta^T x^{(i)}) - g(\theta^T x^{(i)}) + y^{(i)} g(\theta^T x^{(i)}) \right] x_j^{(i)}$$

$$= -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} - g(\theta^T x^{(i)}) \right] \cdot x_j^{(i)}$$

$$\therefore \frac{\partial J}{\partial \theta_j} = -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} - \underbrace{g(\theta^T x^{(i)})}_{h_\theta(x^{(i)})} \right] \cdot x_j^{(i)}$$

$$\boxed{\frac{\partial J}{\partial \theta_j} = -\frac{1}{n} \sum_{i=1}^n \left[h_\theta(x^{(i)}) - y^{(i)} \right] \cdot x_j^{(i)}}$$

2.2 To find the Hessian Matrix of $J(\theta)$

$$\text{ie } \nabla J(\theta) = \frac{\partial}{\partial \theta_k} \left[\frac{\partial J}{\partial \theta_j} \right] = \frac{\partial}{\partial \theta_k} \left[\frac{1}{n} \sum_{i=1}^n [h_{\theta}(x^{(i)}) - y^{(i)}] \cdot x_j^{(i)} \right]$$

$$= \frac{\partial}{\partial \theta_k} \left[\frac{1}{n} \sum_{i=1}^n [h_{\theta}(x^{(i)}) - y^{(i)}] \cdot x_j^{(i)} \right]$$

$$= \frac{\partial}{\partial \theta_k} \left[\frac{1}{n} \sum_{i=1}^n [g(\theta^T x^{(i)}) - y^{(i)}] \cdot x_j^{(i)} \right]$$

$$\nabla J(\theta) = \frac{1}{n} \sum_{i=1}^n g'(\theta^T x^{(i)}) \cdot x_j^{(i)} \cdot x_k^{(i)}$$

derivative
became zero
as no θ is
involved.

$$\nabla J(\theta) = \frac{1}{n} \sum_{i=1}^n g(\theta^T x^{(i)}) \cdot (1 - g(\theta^T x^{(i)})) \cdot x_j^{(i)} \cdot x_k^{(i)}$$

$$\nabla J(\theta) = \frac{1}{n} \sum_{i=1}^n h_{\theta}(x^{(i)}) \cdot (1 - h_{\theta}(x^{(i)})) \cdot x_j^{(i)} \cdot x_k^{(i)}$$

$$H = \nabla J(\theta) = \frac{1}{n} \sum_{i=1}^n h_{\theta}(x^{(i)}) (1 - h_{\theta}(x^{(i)})) \cdot x^{(i)} \cdot x^{(i)}$$

is the Hessian of $J(\theta)$.

2.04 To prove Hessian is positive semidefinite.
ie $z^T H z \geq 0$ for any vector z .

$$\begin{aligned}
 z^T H z &= z^T \left[\frac{1}{n} \sum_{i=1}^n h_0(x^{(i)}) (1 - h_0(x^{(i)})) \cdot x^{(i)} x^{(i)^T} \right] \cdot z \\
 &= \frac{1}{n} \sum_{i=1}^n z^T \cdot \left[h_0(x^{(i)}) \cdot (1 - h_0(x^{(i)})) \cdot x^{(i)} x^{(i)^T} \right] \\
 &= \frac{1}{n} \sum_{i=1}^n [h \cdot (1-h)] z^T x x^T z \\
 &= \frac{1}{n} \sum_{i=1}^n [h \cdot (1-h)] \cdot (z^T x)^2 \\
 z^T H z &= \frac{1}{n} \sum_{i=1}^n [h \cdot (1-h)] \cdot \underbrace{(z^T x)^2}_{\text{is always positive}}
 \end{aligned}$$

h is just a probability, as its squared ie ≥ 0 .
Hence $h(1-h) \geq 0$.

Since both terms are ≥ 0 .

we can say $z^T H z \geq 0$ for any vector z

Hence the Hessian matrix H of $J(\theta)$
is a positive semidefinite.